

CEP-RF — Phase 3

Data Preprocessing & ML-Ready Dataset Preparation

CEP-RF Project | Phase Description

1. Phase 3 in brief

Phase 3 converts the audited CIC-DDoS2019 raw data into a clean, reproducible, machine-learning-ready representation. The phase begins from the authoritative 65 variable numeric candidate features established by Phase 2. It applies only preprocessing decisions justified by the audit and deliberately postpones correlation-based feature reduction and model-driven feature selection until after the all-valid-features Random Forest baseline.

Component	Phase 3 objective
Input	17 CIC-DDoS2019 Parquet files from the raw dataset
Target	`Label` as the authoritative target column
Constant features	Remove the 12 globally zero-variance features identified in Phase 2
Feature set	65 variable numeric features as the baseline starting set
Correlation	33 high-correlation pairs retained for later feature-selection experiments
Protocol	Retained as a legitimate network feature
Missing / invalid values	Handle NaN and $\pm\text{Inf}$ consistently and reproducibly
Scaling	Not required for Random Forest; avoid unnecessary transformation
Raw data	Never modify the original Parquet files
Output	Clean, reproducible ML-ready train/test data plus metadata

2. Phase 3 starting point

Phase 2 established 78 columns in each file: one target column (`Label`) and 77 numeric candidate features. Twelve features were found to be globally constant at 0.0, leaving 65 variable numeric candidate features. No near-constant feature/file pairs were identified, and no confirmed leakage was found. `Protocol` was explicitly allowlisted and retained.

3. Preprocessing workflow

Raw CIC-DDoS2019 → schema validation → remove the 12 global constants → separate `Label` from the feature matrix → normalize numeric representation → handle NaN/Inf → preserve train/test separation → validate the processed feature matrix → save processed data and preprocessing metadata.

4. Constant-feature removal

These 12 features are removed before the baseline because each has a single value (0.0) and zero variance across the audited dataset. They cannot discriminate between classes and therefore provide no useful predictive information.

1. Bwd PSH Flags
2. Fwd URG Flags
3. Bwd URG Flags

4. FIN Flag Count
5. PSH Flag Count
6. ECE Flag Count
7. Fwd Avg Bytes/Bulk
8. Fwd Avg Packets/Bulk
9. Fwd Avg Bulk Rate
10. Bwd Avg Bytes/Bulk
11. Bwd Avg Packets/Bulk
12. Bwd Avg Bulk Rate

5. Target preparation

The `Label` column is treated as the authoritative target. Dataset filenames are not used as the class definition because Phase 2 found that individual files can contain both benign and attack labels. The preprocessing stage therefore separates the target from the feature matrix and records the target encoding used by the implementation.

6. Numeric data cleaning

The preprocessing pipeline checks numerical features for missing values and invalid infinite values (`NaN`, `+Inf`, and `-Inf`). Handling must be deterministic and documented so that training and testing data are treated consistently without using test-set information to fit preprocessing decisions.

7. Train/test separation and leakage control

The existing CIC-DDoS2019 train/test organization is preserved. Preprocessing decisions must not be fitted using test data. The test portion remains unseen for later model selection and final evaluation. This is essential because the project intends to compare the all-valid-features baseline with later feature-selection experiments.

8. What Phase 3 does not do

- It does not remove the 33 high-correlation feature pairs.
- It does not perform feature-importance-based selection.
- It does not apply PCA or arbitrary dimensionality reduction.
- It does not tune the Random Forest.
- It does not modify the raw Parquet files.
- It does not use the test set to choose preprocessing or features.

9. Output of Phase 3

The completed phase should provide a reproducible ML-ready dataset and metadata. The metadata should record the input files, original feature count, removed constants, final feature count and feature order, target column and label mapping, missing/Inf handling decisions, and train/test row counts.

10. Transition to Phase 4

Phase 3 produces the controlled starting point for model experiments. The first experiment is the all-valid-features Random Forest baseline using the 65 variable numeric features. The 33 correlation pairs remain available for later feature-selection experiments so that performance and efficiency can be compared fairly against the baseline.

Phase 3 status

Objective: Produce a clean, reproducible ML-ready dataset from the Phase 2 audited feature set.

Starting feature count: 65 variable numeric features.

Next experimental stage: All-valid-features Random Forest baseline.